Indian Institute of Technology Guwahati
Department of Electronics and Electrical Engineering
EE101 - Basic Electronics

Tutorial - 8 **Date: Oct 12, 2023**

1. A half-wave rectifier with a Si diode having a cut-in voltage of 0.7 V is used to supply 50 V DC to a resistance load of 800 Ω. Calculate the peak value of a sinusoidal AC input required for the same.

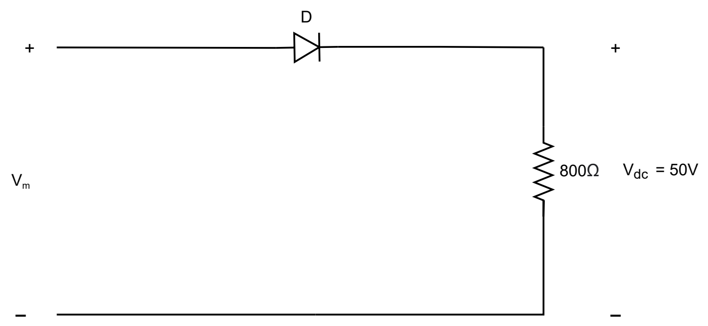

Figure 1:

2. Plot the voltage across the load R_L for the given input. Assume that the time constant RC of the circuit is large compared to T, the time period of the signal. The diodes are considered ideal.

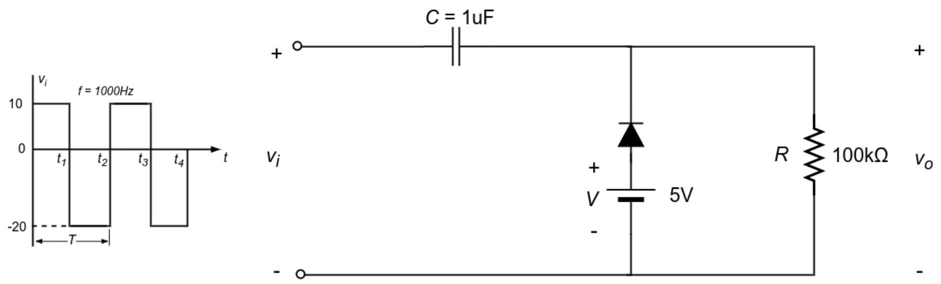

Figure 2:

3. For the Zener diode network, determine V_L, V_R,I_Z and P_Z for (a) $R_L = 1k\Omega$ and (b) $R_L = 3k\Omega$

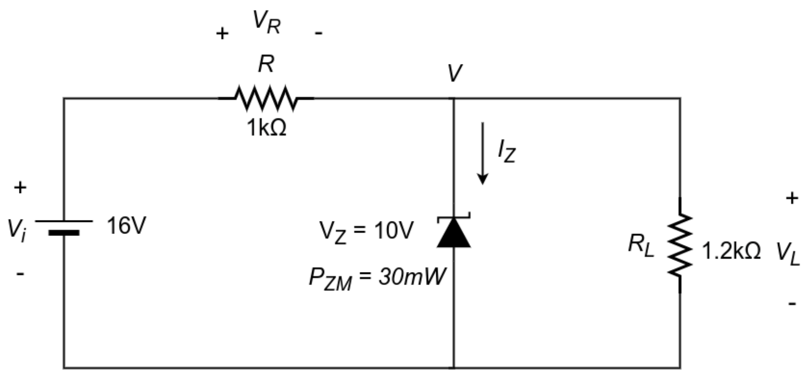

Figure 3:

4. Analyze the system shown in the Figure below and determine the power delivered to each of the three loads as well as the power lost in the neutral wire and each of the two lines.

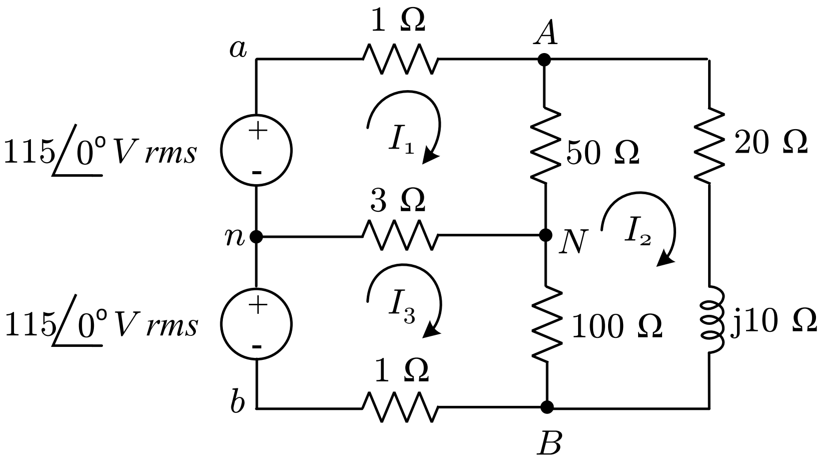

Figure 4: A typical single-phase three-wire system.

5. For the circuit shown in the figure below, find the average power supplied by the source and the average power absorbed by the resistor.

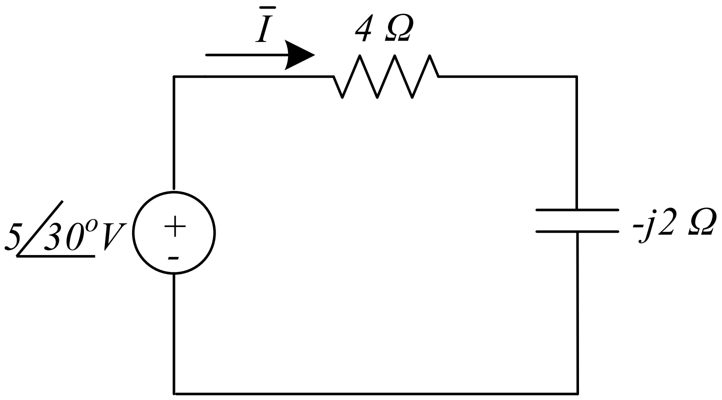

Figure 5:

6. A load $\overline{Z}$ draws 12 kVA at a power factor of 0.856 lagging from a 120 V (RMS) sinusoidal source. Calculate (A) the average and reactive powers delivered to the load, (B) the peak current, and (C) the load impedance.

That's it.